Probabilistic Robotics

Inspired by *Probabilistic Robotics*

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Contents

1 Introduction **2**
1.1 Why uncertainty is unavoidable . . . 2
1.2 Represent a belief, not just a best guess . . . 2
1.3 Global localization: the hallway example . . . 3
1.4 Uncertainty should influence action . . . 4
1.5 Benefits and costs of the probabilistic approach . . . 4
1.6 How the ideas fit together . . . 5

Chapter 1: Introduction

A robot must perceive and act in a physical world it does not fully know. The central idea of probabilistic robotics is to represent that uncertainty explicitly and use it when interpreting observations and choosing actions. The aim is reliable behavior despite incomplete information, rather than pretending that a single estimate is certainly correct.

1.1 Why uncertainty is unavoidable

A factory can constrain where objects are placed and how a task is performed. A home or road is less controlled: people move, objects change, and unexpected situations arise. Five sources of uncertainty matter even when the software executes exactly as intended.

1. **The environment.** The world changes in ways the robot cannot fully predict, such as a person stepping into its path.
2. **Sensors.** Limited range and resolution hide information; measurement noise perturbs what is visible. A camera cannot see through a wall, and seeing a door does not identify which of several similar doors it is.
3. **Actuation.** A commanded movement is not an exact displacement. Wheel slip, control noise, and mechanical wear change the outcome.
4. **Models.** A model simplifies the underlying physics. Even a perfect calculation with an imperfect model can give an inaccurate prediction.
5. **Computation.** A robot must respond on time. An approximation that finishes before the next decision may be more useful than an exact calculation that arrives too late.

Better hardware can reduce some errors, but it cannot eliminate missing information, model simplifications, or ambiguity. Robust software must account for these limitations.

1.2 Represent a belief, not just a best guess

Definition 1.1 [Belief]. A belief is a probability distribution over possible states of the world, given the robot's information. For localization, the hypotheses are possible robot locations in a known map. Write this distribution as $\text{bel}(x)$.

In a discrete model, probabilities sum to one. For a continuous position, bel(x) is a density: probability is area under its curve, not the curve's height at one point. A belief can have several separated peaks, preserving different explanations of the same sensor reading.

For example, a robot may be either near the kitchen or near the bedroom because the two doorways look alike. Choosing one location immediately loses the alternative. Keeping both hypotheses lets later observations resolve the ambiguity. This uncertainty is different from a small error around one known doorway; one narrow peak cannot describe both situations.

1.3 Global localization: the hallway example

Localization estimates a robot's position in an external reference frame using sensor data and a map. In global localization, its starting location is unknown. Consider a hallway with three indistinguishable doors.

1. **Start with a prior.** With no reason to favor one position over another, use a uniform belief over the allowed hallway positions.
2. **Observe a door.** Locations near any of the three doors become more plausible. Other positions retain some probability if the detector can make a false report. One observation does not select a unique door.
3. **Move.** Shift the hypotheses by the estimated displacement and spread them to account for motion error. Uncertain motion generally broadens the possible current locations.
4. **Observe another door.** Favor hypotheses consistent with both the movement and the new reading. If only one door-to-door spacing fits well, one peak becomes dominant. In a repetitive hallway, ambiguity may remain.

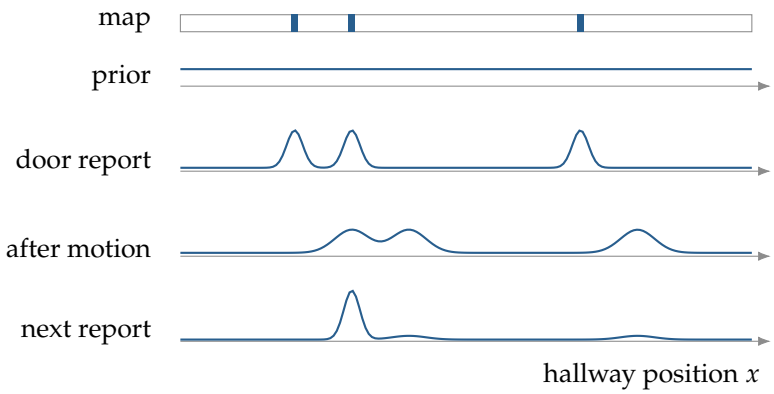


Figure 1: A schematic belief sequence. Motion shifts the three hypotheses to the right and broadens them; the next door report favors one. Heights have no numerical scale, and the rows are not a computed filtering run.

1.3.1 A small numerical illustration

Take four possible positions: three door locations A, B, C and a non-door location N. Give each prior probability $1/4$. Suppose a detector reports "door" with probability $9/10$ at a door and $1/10$ at N. Reweight each prior by the likelihood of the report:

position	A	B	C	N
prior	$1/4$	$1/4$	$1/4$	$1/4$
likelihood	$9/10$	$9/10$	$9/10$	$1/10$
product	$9/40$	$9/40$	$9/40$	$1/40$
posterior	$9/28$	$9/28$	$9/28$	$1/28$

The products sum to $28/40$, so dividing by that sum produces a distribution. The report strongly favors being at *some* door, but each individual door still has probability only $9/28$. It also leaves a small chance that the report was wrong. These invented numbers illustrate the reweighting; the general recursive Bayes filter is developed in Chapter 2.

1.4 Uncertainty should influence action

Perception estimates what is happening; action changes the world and what can be learned about it. A robot uncertain between two locations may move toward a distinctive landmark before committing to a route. That action has an information benefit even if it does not immediately shorten the journey. This creates a tradeoff between EXPLORATION, gathering information, and EXPLOITATION, using current knowledge to accomplish the task.

A decision should therefore use the relevant belief, not just its most likely state. Probabilistic representation alone does not select an action: a task objective and a model of possible consequences are also needed.

1.5 Benefits and costs of the probabilistic approach

Explicit uncertainty gives a common language for combining imperfect models with noisy measurements. It supports robust localization, recovery after a robot is moved unexpectedly, and map building without a reliable global position reference. Learning an unknown model can likewise be viewed as estimating persistent unknown quantities from accumulated data.

These are capabilities of suitable algorithms, not guarantees from the word "probabilistic." A model that rules out the true location cannot recover it merely by normalizing its remaining hypotheses. Incorrect assumptions or unjustifiably small uncertainty can still produce confident mistakes.

The main cost is representing and updating a distribution. A general density over a continuous state space cannot be stored exactly in a finite table. Compact representations such as Gaussians,

discretizations, and sample-based approximations trade expressive power against memory and computation. A single Gaussian is compact, for example, but cannot preserve three distinct door hypotheses. The representation must fit both the uncertainty and the available computation.

1.6 How the ideas fit together

Classical approaches emphasize planning with models; reactive approaches respond directly to sensing; hybrid systems combine fast reactions with higher-level planning. The probabilistic approach combines models and perception through explicit uncertainty at either level. Randomized search is a separate idea: an algorithm can use random numbers to search a known world without representing uncertainty about that world. Probabilistic estimation also has roots in Kalman filtering and early joint localization and mapping methods.

The progression is from state-estimation filters to motion and sensor models, then localization and mapping, and finally action selection under uncertainty. Understanding an algorithm requires its derivation, executable procedure, implementation limitations, and experimental behavior. A correct equation alone does not establish that a robot implementation behaves reliably.

Source. Sebastian Thrun, Wolfram Burgard, and Dieter Fox, Probabilistic Robotics, early draft, Chapter 1, pp. 1–8 (PDF pp. 13–20).